Original Paper

European Neurology

Eur Neurol 2010;64:140–144
DOI: 10.1159/000316764

Received: February 8, 2010
Accepted: June 7, 2010
Published online: July 29, 2010

Acute Stroke Patients Have Occult Malignancy More Often than Expected

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Key Words
Acute ischemic stroke · Occult malignancy · Nonbacterial thrombotic endocarditis

Abstract
The aim of the present study was to investigate the frequency of having occult malignancy in patients with acute ischemic stroke and their clinical characteristics. We retrospectively enrolled 1,714 consecutive ischemic stroke patients within 7 days of onset. The patients were divided into two groups: the Non-M group had no malignancy, and the M group had malignancy. We compared the clinical characteristics of the two groups. Of 1,714 ischemic stroke patients, 51 patients (3.0%; M group) were newly diagnosed as having malignancy. The M group was significantly older than the Non-M group ($p = 0.009$). Hemoglobin (Hb) was less and D-dimer was higher in the M group than in the Non-M group ($p < 0.001$). The patients with both D-dimer ≥1.3 ng/dl and Hb <12.8 g/dl more frequently had occult malignancy than patients without ($p = 0.0088$). Copyright © 2010 S. Karger AG, Basel

Introduction

Cancer and stroke are the second and third leading causes of death, respectively, in Japan. Both disorders increase in incidence and prevalence with age. In patients with cancer, stroke frequently occurs. Graus et al. [1] found evidence of cerebral infarction in 7% of 3,426 autopsied cancer patients. Nonbacterial thrombotic endocarditis (NBTE) is the most common cause of stroke in cancer patients [2–4]. It is thought to occur via a state of chronic disseminated intravascular coagulation (DIC) that predisposes cardiac valves to edema, degeneration and deposition of sterile platelet-thrombin vegetation [5]. Cerebral intravascular coagulation is the second most common mechanism of stroke in cancer patients. However, such a mechanism of stroke frequently occurs in the terminal stage of malignancy [6–8].

Cancer-related stroke can occur at any point during the course of malignancy [9]. In fact, in a few patients, stroke is the first manifestation of occult malignancy. If occult malignancy in patients with acute stroke could be identified and treated, the patients' prognosis should improve. Therefore, it is important to search for occult malignancy in acute stroke patients. The aim of the present study was to investigate the frequency of occult malignancy in patients with acute ischemic stroke and their clinical characteristics.

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0014–3022/10/0643–0140$26.00/0

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Participants and Methods

Acute ischemic stroke patients admitted to our stroke center (Stroke Center, Kawasaki Medical School, Kurashiki, Japan) within 7 days of stroke onset between March 2004 and August 2009 were retrospectively studied. All patients had a baseline clinical assessment on admission with ECG, chest X-ray, abdominal X-ray, and brain CT or MRI, including diffusion weighted imaging, fluid-attenuated inversion-recovery images, T2*-weighted imaging and MR angiography. All patients underwent duplex carotid sonography to evaluate stenotic carotid lesions. Blood samples for measurement of hemoglobin (Hb), WBC, D-dimer, fibrinogen, AST, ALT and C-reactive protein were obtained at baseline.

Information about patient history, including cardiovascular risk factors (hypertension, diabetes mellitus, hyperlipidemia and smoking), atrial fibrillation (AF) and malignancy history, was obtained. Cardiovascular risk factors were defined as follows: (1) hypertension, use of antihypertensive agents, systolic blood pressure >160 mm Hg or diastolic blood pressure >95 mm Hg on admission; (2) diabetes mellitus, use of oral hypoglycemic agents or insulin or glycated Hb >6.4%; (3) hyperlipidemia, use of antihyperlipidemic agents or serum cholesterol level >220 mg/dl; and (4) smoking or any lifetime cigarette use. Patients with previous malignancy that had been treated and from which they had recovered completely were excluded from this study. Thus, only stroke patients who had not previously been diagnosed as having cancer were enrolled.

Using Trial of ORG 10172 in Acute Stroke Treatment criteria [10], ischemic stroke subtypes were categorized into five groups: (1) large-artery atherosclerosis; (2) cardioembolism; (3) small-artery occlusion (lacune); (4) stroke of other determined etiology; and (5) stroke of undetermined etiology. Diagnoses are based on clinical features and data collected by tests such as brain imaging (CT/MRI), cardiac imaging (echocardiography, Holter ECG, etc.), duplex imaging of extracranial arteries, angiography, contrast-transcranial Doppler examination and laboratory assessments for a prothrombotic state.

Transesophageal echocardiography (TEE) was done to detect right-to-left shunt (RLS) and NBTE [2–4, 11]. Venous ultrasonography of the lower limb was carried out to detect deep venous thrombus.

Malignancy Survey

Patients had fecal occult blood tests, chest X-rays and laboratory data as routine tests. If patients had positive fecal occult blood tests, a search for malignancy was conducted using tumor markers (CEA, carbohydrate antigen 19-9, prostate-specific antigen for men, carbohydrate antigen 125 for women), abdominal echography, abdominal CT, gastroscopy and colonoscopy. If patients had suspected lung cancer on chest X-ray, contrast chest CT was performed. When malignancy was diagnosed, it was evaluated using the TMN staging system.

Statistical Methods

All patients were divided into two groups: the no malignancy group (Non-M group) and the newly diagnosed malignancy group (M group). All data were entered into SPSS software (Version 13.0 for Windows) for storage and analysis. Baseline characteristics and laboratory data were compared between the two groups using the Mann-Whitney U test and the χ^2 test. If there were significant differences in continuous variables between the Non-M and M groups, the optimal cutoff points of each continuous variable to discriminate the Non-M group from the M group were obtained by receiver operating characteristic (ROC) curves. Differences were considered statistically significant at $p < 0.05$.

Table 1. Baseline characteristics and laboratory data for the two groups

	Non-M n = 1,663	M n = 51	p
Age, years (median)	73	77	0.009
Male, n (%)	1,044 (63)	30 (59)	0.565
TIA, n (%)	252 (15)	11 (22)	0.211
Ischemic stroke, n (%)	1,411 (85)	40 (78)	0.211
Small-vessel disease	243 (15)	4 (8)	0.157
Cardioembolic stroke	440 (26)	16 (31)	0.434
Large-vessel disease	130 (8)	5 (10)	0.604
Other determined etiology	54 (32)	5 (10)	0.011
Undetermined etiology	544 (33)	10 (20)	0.048
Hypertension, n (%)	1,115 (67)	32 (63)	0.492
Hyperlipidemia, n (%)	506 (30)	13 (25)	0.431
Diabetes mellitus, n (%)	443 (27)	13 (25)	0.858
Smoking, n (%)	777 (47)	16 (31)	0.029
AF, n (%)	346 (21)	15 (29)	0.142
RLS/TEE, n (%)	401/1,020 (39)	17/30 (57)	0.056
Hb (median), g/dl	13.4	12.1	<0.001
WBC (median), /μl	6,430	6,390	0.925
CRP (median), mg/dl	0.13	0.21	0.499
AST (median), IU/l	21	20	0.327
ALT (median), IU/l	17	18	0.535
Fibrinogen (median), mg/dl	254	293	0.662
D-dimer (median), ng/dl	0.9	1.6	<0.001
Fecal occult blood test, n (%)	232/578 (40)	26/48 (54)	0.058

Mann-Whitney U test, χ^2 test.

Results

A total of 1,822 consecutive ischemic stroke patients were admitted to our stroke center between March 2004 and August 2009. Of these, 108 patients had a history of a malignancy for which they had been treated and recovered completely. Thus, a total of 1,714 consecutive patients (1,074 male; mean age 71.5 years) were enrolled in the present study.

During hospitalization, 51 patients (3.0%; M group) were newly diagnosed as having occult malignancy. Table 1 shows the type of cancer. The cancers were located in the colon in 13 patients, in the stomach in 10, in the lung in 6, in the liver in 4, in the pancreas in 4, in the bile duct in 4, in the prostate in 4, in the kidney in 3, in the uterus in 3, in the breast in 1 and in the ovary in 2. The

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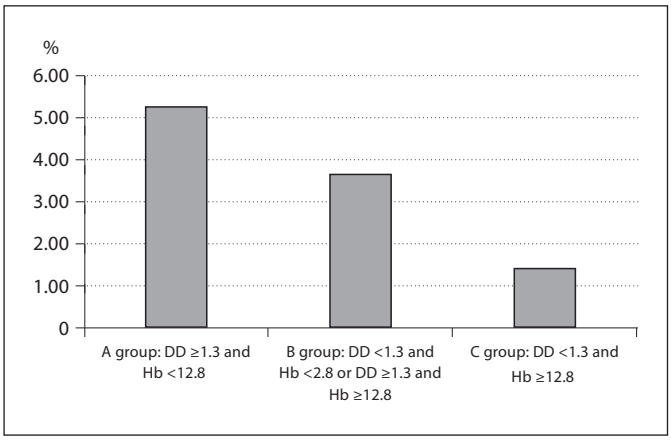

Fig. 1. The frequency of occult malignancy in the three subgroups.

Table 2. Type of cancer in the M group (n; total n = 51)

Colon	13
Liver	4
Pancreas/bile duct	4
Kidney	3
Bladder	1
Prostate	4
Stomach	10
Lung	6
Uterine	3
Breast	1
Ovary	2

histological types were as follows: adenocarcinoma (n = 42), squamous cell carcinoma (n = 5) and hepatocellular carcinoma (n = 4). The diagnostic modality that most often revealed cancer was fecal occult blood testing (n = 20), followed by abdominal echography (n = 10), abdominal CT (n = 10), chest CT (n = 6), prostate-specific antigen (n = 4) and chest X-ray (n = 1).

Table 1 shows the clinical characteristics of the two groups. The M group was significantly older than the Non-M group (77 vs. 73 years; p = 0.009). The M group less frequently had other determined and undetermined etiology than the Non-M group (54 vs. 5; p = 0.011, 544 vs. 10; p = 0.048). Risk factors excluding smoking were not different between the two groups. Hb was less and D-dimer was higher in the M group than in the Non-M group (12.1 vs. 13.4 g/dl, $p < 0.001$; and 1.6 vs. 0.9 ng/dl, $p < 0.001$). The normal range of Hb value has a gender gap: 13.5–17.5 g/dl for men, 11.5–15.0 g/dl for women. Taking gender differences into account, Hb was less in the M group than in the Non-M group (men: 14.1 vs. 13.5 g/dl, p = 0.024; women: 12.6 vs. 11.1 ng/dl, $p < 0.001$). Of 626 patients (37%) assessed by fecal occult blood testing, 258 (41.2%) had positive fecal occult blood. Of these, 26 patients (10.1%) had malignancy. TEE was performed in 1,050 patients (61%). No vegetations were detected, and no patients had NBTE. RLS was higher in the M group than in the Non-M group on TEE (57 vs. 39%, p = 0.056), but the difference was not significant.

Using ROC curves, the optimal cutoff points of Hb and D-dimer to discriminate the Non-M group from the M group were 12.8 g/dl and 1.3 ng/dl, respectively. The patients were divided into three subgroups according to D-dimer ≥ and <1.3 ng/dl and Hb ≥ and <12.8 g/dl: A group, both D-dimer ≥1.3 ng/dl and Hb <12.8 g/dl; B group, both D-dimer <1.3 ng/dl and Hb <12.8 g/dl or both D-dimer ≥1.3 ng/dl and Hb ≥12.8 g/dl; C group, both D-dimer <1.3 ng/dl and Hb ≥12.8 g/dl. The frequency of the presence of occult malignancy was 5.3% (19/361) in the A group, 3.9% (23/592) in the B group and 1.4% (12/841) in the C group ($p < 0.001$; fig. 1). Therefore, the patients with both D-dimer ≥1.3 ng/dl and Hb <12.8 g/dl more frequently had occult malignancy than patients without both [19/361 (5.3%) vs. 35/1398 (2.4%); p = 0.0088].

Discussion

The present study demonstrated that the frequency of occult malignancy diagnosed during hospitalization was at least 3.0%. The incidence of colorectal cancer has been reported to be 16 per 10,000 person-years for the general population of Japan [12]. In the present study, colon cancer was newly diagnosed in 13 patients (0.76%), for an incidence that was about 500 times that for general examinations in Japan. Thus, we believe that the prevalence of occult malignancy was higher than expected in acute stroke patients.

Patients with malignancy had elevated D-dimer and lower Hb levels [13–21]. The present results demonstrated that patients with D-dimer ≥1.3 ng/dl and Hb <12.8 g/dl on admission more frequently had malignancy than those without both. Explanations for the decrease in Hb level in the M group are as follows. First, some of the patients with malignancy had cancer cachexia. Second, some patients had gastrointestinal cancers and bleeding. In fact, 100% of patients with gastrointestinal cancer had positive fecal occult blood tests.

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Previous reports showed that D-dimer levels were associated with NBTE and DVT in cancer patients [17–21]. Edoute et al. [18] reported that D-dimer levels increased in patients with NBTE. However, no patients had NBTE in the present study. Almost all patients with cancer show laboratory evidence of hypercoagulability. Naschitz et al. [5] reported that occult cancer promotes plaque rupture or erosion, releases microparticles from platelets and enhances thrombin formation. Thus, our patients with malignancy had elevated D-dimer levels.

In the present study, colon cancer was the most frequent cancer, followed by stomach cancer. Chaturvedi et al. [6] reported that in 33 patients with cancer-related stroke (Trousseau's syndrome), lung cancer was observed in 9 (27%) and gynecological cancer was observed in 7 (21.2%). Cestari et al. [7] reported that in stroke patients with cancer, lung (30%), brain (9%) and prostate (9%) were the most common cancers. However, these reports included patients who had been treated before stroke onset, but not patients with newly diagnosed occult malignancy after stroke onset. Therefore, the cancer lesions differed between the present and previous studies. Furthermore, in acute ischemic stroke, antiplatelet and anticoagulation therapy was started immediately after admission. Therefore, fecal occult blood tests may likely be positive in patients with potential cancers of digestive organs, such as colon cancer.

The M group demonstrated a trend toward higher rates of RLS ($p = 0.056$). Iguchi et al. [22] reported that RLS was more frequent in patients with malignancy than in patients without malignancy. Paradoxical brain embolism should be considered an important mechanism in patients with stroke with malignancy. We believe that the stroke mechanism of some patients with occult malignancy was paradoxical brain embolism. Taccone et al. [23] reported that acute stroke patients having occult malignancy were 52.4 years old and the most frequent neoplasms were lung and breast cancer. Rogers [24] reported that NBTE and DIC may cause the cerebral infarction in a hypercoagulable state. In our study, the median age of the M group was 77, and the most frequent neoplasms were colon and stomach cancer, NBTE was not found out at TEE and only three patients had DIC. Because many patients were assessed by fecal occult blood testing, abdominal echography and abdominal CT, most frequently neoplasms were tumors of digestive organs. The cancers were detected earlier, and early stage cancers were 57% in the M group. Therefore, the median age of the M group was higher, and NBTE and DIC were less detected.

The present study had some limitations. First, this was a retrospective study, and obstetrical and gynecological surveys were not conducted. Second, after hospital discharge, patients without malignancy were not examined. Thus, the frequency of malignancy might be greater than reported. Third, the methylenetetrahydrofolate reductase (MTHFR mutations) [25, 26], folic acid deficiency [27], Leiden factor V [25, 26] and thrombin gene mutations were not examined in the M group.

In conclusion, at least 3.0% of acute ischemic stroke patients had occult malignancy. The presence of both D-dimer $\geq$1.3 ng/dl and Hb $<$12.8 g/dl on admission appears to be a good predictor for malignancy in acute stroke patients.

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